RateController User Manual

RateController (RC) is a Windows application for controlling the application rate of agricultural products (liquids, granules, seeds, ammonia) in conjunction with AgOpenGPS. It communicates with rate control modules via UDP/Ethernet to manage product application rates, boom sections, and tank quantities.

RC can also be used without AgOpenGPS. Automatic rate control based on ground speed will continue to work normally. Section control will be manual only.

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1. Getting Started

Before operating, complete these setup steps in order:

1. **Create or load a Profile** — a profile stores all machine settings (products, sections, relays, etc.)
2. **Connect the board** — plug it into the same network as this PC. Go to **Modules > Communication** and set the subnet. An ESP32 on WiFi is connected from **Modules > Communication > WiFi** instead
3. **Check the board's firmware** — each board needs the firmware released with this version of RateController. A board with older firmware shows **Update firmware** on Modules > Boards, or isn't listed at all. See Updating Module Firmware
4. **Assign a module ID** — go to **Modules > Boards**, select the **New** board, choose a module ID (0–7), press Save, and accept the starting configuration for its type. See Setting Up a New Board
5. **Configure the module** — go to **Modules > Config** to check the relay type and sensor count
6. **Configure pins** — go to **Modules > Pins** to verify or set sensor and PWM pin assignments
7. **Configure relay pins** — go to **Modules > Relay Pins** to assign physical pins to each relay
8. **Set valve mode** — go to **Modules > Valves** to select 2-wire or 3-wire valve control
9. **Define sections** — go to **Machine > Sections** and set section widths
10. **Configure relays** — go to **Machine > Relays** and assign relay types to sections
11. **Configure switches** — go to **Machine > Switches** to set master switch behavior
12. **Set up your product** — go to **Products > Rate** and enter product name, control type, units, and rates
13. **Assign the sensor** — go to **Products > Settings** to assign the module ID and sensor ID
14. **Calibrate** — go to **Machine > Calibrate** to determine the Cal Factor
15. **Create a Job** — go to **File > Jobs** to create a job for the field
16. **Connect AgOpenGPS** — the AOG status indicator should turn green

Using Rate Control with AgOpenWeb

- Run RC and AgOpenWeb on the same PC, and close AgOpenGPS and AgIO.
- Open a field in AgOpenWeb. RC only gets speed and section states while a field is open.
- Section changes reach RC when you press Save in AgOpenWeb, or 10 seconds after your last edit.
- RC's switches control AgOpenWeb's sections, as they do with AgOpenGPS.

Data Folder (File > Folders)

RC stores its data — profiles, jobs, fields, rate maps, and logs — in a data folder (Documents by default). Use **File > Folders** to move it, e.g. to a synced folder (OneDrive, Dropbox) or another drive.

Control	Description
Current location	The data folder currently in use
New location	Press to browse for the new data folder; the chosen path can also be typed in the box
Copy Data	Copy the contents of the current data folder to the new location. Leave unchecked to simply switch to a folder that already contains RC data.
Overwrite	Allow copying into a folder that already contains files. Without this, Copy Data only copies into an empty folder.

Press **OK** and confirm — RC changes the folder and restarts. The new location cannot be a sub-folder of the current data folder.

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2. Profiles

Menu: File > Profiles

A profile stores all machine configuration settings. Multiple profiles can be maintained for different implements.

Button	Action
1. New	Create a new profile
2. Load	Load the selected profile
3. Delete	Delete the selected profile. Default, and the last remaining editable profile, can not be deleted
4. Copy	Copy the selected profile to a new name. The copy is always editable, even when copied from Default
5. File name	Enter name for New or Copy
6. Files	List of available profiles to select
7. Page Up/Down	Scroll through the profile list

The Default Profile

Default is a read only reference profile. It holds a known good starting configuration and is never changed by the program, so it is always available to start over from. It can not be deleted.

Because it is read only, settings changed while Default is loaded are **not saved**. To warn you of this, the profile name at the top left of the menu is shown on a **red background** whenever a read only profile is loaded.

To start a new setup from the reference configuration, select **Default**, type a name, and press **Copy**. The copy is a normal editable profile and is loaded automatically.

> Because Default can not be saved to, the program always keeps one editable profile. On first run it makes one from Default and names it **Example**. Example is an ordinary profile — edit it, copy it, or delete it once you have a profile of your own. It is only rebuilt if no editable profile is left.

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3. Jobs

Menu: File > Jobs

Jobs track application activity per field. Each job stores application data, notes, and a rate map.

Current Job

Field	Description
Name	Job name
Date	Date/time of the job
Field	Select from list or type a new field name
Notes	Free-text notes for the job

Jobs List

The list shows each job's **Name**, **Date**, and **File**. Click a job to select it.

Control	Action
Year	Year used by the filter
Field	Field used by the filter
Filter	When checked, only jobs matching the selected Year and Field are listed
 Field	Delete the selected field from the Fields list (not allowed while the field is in use)
 Load	Open the selected job
 New	Create a new job
Copy	Copy selected job and its rate map to a new file
 Delete	Delete selected job
Show at Start-Up	Display the Jobs menu when the app launches
 Activity	Job report
 Import	Import jobs
 Export	Export all jobs to a chosen folder

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4. Options – Speed Source, Display & Phone

Menu: File > Options

The page has two tabs. **Speed** selects the source of ground speed used for rate calculations and configures display units. **Phone** connects a phone to RateController's phone page (see Phone Tab below).

Speed Source

Option	Description
GPS	Use speed from AgOpenGPS
Simulated Speed	Simulate a constant speed (useful for testing)
Wheel Sensor	Use a speed sensor connected to a module GPIO pin

Wheel Sensor Setup

1. Enter the **Module ID** and **GPIO pin** used for speed sensing.
2. Enter the calibration number in **pulses per mile** or **pulses per kilometer**.
3. Press the **up arrow** to send configuration to the module (also sent on Save).
4. Use the **Calibration screen** to calculate the calibration number if unknown.

Display Options

Option	Description
Metric	Toggle between metric (km, ha, L) and imperial (mi, ac, gal) units
Compact Display	When on, the arrow on the main screen collapses it to just the rates instead of minimizing it to the small restore window. Tap the collapsed display to expand it again.

The rate display window is no longer switched on here. Each product now has its own **Popup Display** button on Products > Settings, so a separate window can be shown for each product.

Phone Tab

RateController has a built-in page for a phone's web browser, so the switches and calibration can be used away from the tablet: walking the boom to check nozzles, or standing at the meter with a catch bag. It is always available; there is nothing to install on the phone and nothing to switch on in RateController.

Connecting a Phone

1. Put the phone and the tablet on the **same network**. Any of these works:
    - **The phone's hotspot**: connect the tablet to it.
    - **A WiFi router** (Starlink, a cellular router or any other): connect both to its main network.
    - **The tablet's own hotspot**: choose **Tablet Hotspot** on Modules > Communication > WiFi and press Save, then connect the phone using the network name and password shown on this tab.

2. **Phone hotspot or router only:** once the tablet is connected, tap **WiFi settings**, tap the network and set it to **Private**. Windows blocks the phone page on a Public network. Do this again for a new phone or after renaming the phone's hotspot, because Windows treats it as a new network.
3. Scan the code with the phone's camera to open the page. Adding it to the phone's home screen makes it quicker next time.

Phone Tab Controls

Control	Description
QR code	Opens the phone page when scanned. When the tablet is on more than one network, tap the code to show the next address. It updates by itself when a network is joined or left.
Address	The same address as text, for typing into the phone's browser. "(1 of 3, tap for next)" shows when there are several.
Tablet Hotspot	The hotspot's state: On, Off, Checking..., *On, firewall not allowed* or *Off, restarts failed*. The hotspot is switched on and off on Modules > Communication > WiFi; the **Comm → WiFi** link opens that page.
Network / Password	The tablet hotspot's network name and password, for joining it from the phone, or Windows' reason when the hotspot can't start.
WiFi settings	Opens the Windows WiFi page, for setting a network to Private.

Tablet Hotspot

Set on Modules > Communication > WiFi. The same hotspot carries both the phone and any ESP32 modules joined to it.

- The first time Tablet Hotspot is saved, Windows asks for **administrator permission** to let the phone page through the firewall. This is needed once per tablet. If it's declined, the phone can join the hotspot but the page won't open, and the status shows *On, firewall not allowed*. On a standard Windows account the prompt asks for an administrator's password.
- The tablet's hotspot address is always **192.168.137.1**, so a bookmark or home-screen link on the phone keeps working.
- If the hotspot turns off, RateController turns it back on within about 30 seconds, up to 3 times. After that the status shows *Off, restarts failed*. To try again, choose Current WiFi Network on Comm → WiFi and Save, then Tablet Hotspot and Save, or restart RateController.
- Windows' **Power saving** turns the hotspot off when no device has been connected for 5 minutes. If necessary, turn it off in Windows' **Hotspot settings** (linked from Comm → WiFi). RateController can't change this setting.
- Windows shares the tablet's internet connection through the hotspot. If the tablet connects to a router by WiFi as well, its WiFi adapter must be able to do both at once; not all adapters can.

Router Tips

- Use the router's main network, not a guest network, and make sure **AP isolation / client isolation** is off. Either one stops the phone from reaching the tablet.
- A **DHCP reservation** for the tablet in the router keeps its address the same, so the phone's bookmark keeps working.
- The phone has to stay within the router's WiFi range at the boom.

Phone Page

The phone opens a menu titled **RateController** with two buttons: **Switches**, a copy of the on-screen switch panel (see Phone Switches in §14), and **Calibration**, product calibration (see Calibrating from a Phone in §11). Each page has a **Menu** button at the bottom to go back. The page uses the tablet's display colours.

> **Note:** The phone page has no password. Anyone on the same network can open it, so use your own hotspot or network.

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5. Products – Rate

Menu: Products > Rate

Configures the fundamental properties of a product.

Setting	Description
1. Product Name	Name of the product
2. Control Type	How the application rate is controlled (see below)
3. Quantity Units	Units used to measure the product (L, gal, kg, lbs, seeds, etc.)
4. Coverage Units	Acre, Hectare, Minute, or Hour
5. Sensor Counts/Unit	Number of sensor pulses per unit of product applied
6. Density	Product density (used for mass/volume conversion)
7. Base Rate	Target application rate
8. Alt. Rate (%)	Alternate rate as a percentage of the base rate
9. Tank Size	Total tank capacity

Control Types

Selects the actuator driving the product and how the PID loop positions it. This must match the physical valve/motor wiring on the module — an incorrect selection can leave the actuator open or unresponsive.

Type	Description
Standard Valve	A proportional flow-control valve driven continuously by the PID loop — includes a motorized ball valve or a solenoid valve. PWM = 0 stops the valve motor and holds its last position (does not drive it closed) — use this for valves with no dedicated closed/shutoff position.
Combo Close	Same continuous PID control as Standard Valve — includes a motorized ball valve or a solenoid valve — but the valve is also driven fully closed whenever the product/section is off (rather than just holding position). Use for valves that must shut off flow, not just stop adjusting.
Motor	Drives a variable-speed motor that IS the metering device itself (e.g. a hydraulic or electric drive motor turning a metering roller/auger/pump) — not a motorized valve. PID output accumulates over time (integrating control) instead of positioning directly, so it needs different gain tuning than a valve.
Combo Timed	For close-only/on-off style valves that can't be positioned proportionally. Instead of continuous PWM, the valve is pulsed through alternating Adjust/Pause cycles (set on the Control page) to work the rate toward target, with a fast minimum-start ramp from zero.

Fans

The last two products in the list are fans, named Fan 1 and Fan 2. Selecting one replaces the product settings with the fan settings, because a fan is controlled to a speed rather than to an application rate.

Setting	Description
Sensor Counts / RPM	Number of sensor pulses per revolution
Target RPM	The speed the fan is controlled to
Current RPM	Measured fan speed (read-only)
Error %	How far the measured speed sits from the target, as a percentage (read-only)

To put a fan's speed in its own floating window, use **Popup Display** on Products > Settings (see Popup Display Window). The same button serves every product, so a fan window and a product rate window work the same way.

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6. Products – Settings

Menu: Products > Settings

Configures sensor assignment, alarm thresholds, and display preferences per product.

Setting	Description
1. Enable product	Activate this product
2. Rate Sensor Location	Module ID (0–7) and Sensor ID (0–1) for the flow sensor
3. Minimum UPM	Minimum flow rate the controller will maintain. Enter either a fixed **UPM** value or a **By Speed** value (MPH/KPH) — the hint below the boxes shows the equivalent value the other way (e.g. the speed that corresponds to the entered UPM).
4. Off-Rate Alarm	% error in rate that triggers an alarm
5. Default Product	Product displayed at startup
6. Scale Weight	Show the scale weight screen
7. Popup Display	Open a floating window showing this product's rate — or this fan's speed (see below)

Popup Display Window

Popup Display opens a small floating window with the live value for that product: the applied rate, or the speed if the product is one of the two fans. Press Save to apply it. Every product has its own button and its own window, so several can be on screen at once; each window is labelled with the product it belongs to. The windows have no title bar and use the display colours set under Options.

Click and drag a window (either mouse button) to reposition it. Where it is dropped decides what it follows. Dropped overlapping the main screen it follows the main screen. Dropped overlapping another floating window — another popup display, the pressure display, or the on-screen switch panel — it follows that window instead, so a group of windows can be built up and moved together. If it overlaps both, the main screen wins. Dropped clear of everything it follows nothing until it is dropped onto something again.

A window detaches while the main screen is minimized and re-attaches when the main screen is restored. If the window it was following is closed, it stays where it is and simply stops following. Each window's position is remembered separately between sessions.

A product that is not enabled has no display window, whatever this button is set to, and enabling the product again brings its window back.

Controls

Control	Action
Navigate left	Previous product
Navigate right	Next product
Cancel	Discard changes
Save	Save settings

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7. Products – Mode

Menu: Products > Mode

Sets the operating mode for each product. Navigate between products with the left/right arrows.

Mode	Description
Controlled UPM	PID feedback control — adjusts motor/valve to maintain the target rate as speed changes
Constant UPM	Maintains a fixed flow rate based on total machine width regardless of which sections are on. Used for sprayers where turned-off sections return flow to the tank (after the flow meter). RC corrects the measured flow by subtracting the estimated contribution of the off sections to give an accurate application rate per acre/hectare.
Document Applied	Tracks and records material that has already been applied — no active control
Document Target	Tracks and records the target rate for comparison — no active control
Document Applied Manual Rate	For implements with a fixed mechanical/manual metering that has no flow sensor or module at all (e.g. a fertilizer spreader gate set to a position). Enter the implement's flow output with all sections on, in units per minute, in the Manual Rate box; RC computes the displayed/logged application rate from that fixed value and ground speed, and scales the flow down when sections are switched off — no active control, no module.

Manual Rate (enabled only when Document Applied Manual Rate is selected): enter the implement's measured output in units per minute, **not** per acre/hectare — typically found by running the implement at its set position and weighing or measuring output over a timed interval. Because the value is a constant flow, the displayed/logged rate per acre/hectare will rise and fall with ground speed even though the implement setting hasn't changed — this is expected for fixed-metering implements.

Enter the output of the **whole machine with every section running** — on an implement with more than one dispenser (two gates, two spinners, and so on) that is their combined output, not one dispenser's. When a section is switched off RC lowers the reported flow by that section's share of the implement width, so with two equal dispensers and one shut off the flow halves; the ground being covered halves at the same time, so the rate per acre/hectare correctly stays where it was.

This requires **one section per dispenser**, with each section's width set to the ground that dispenser actually covers. There is no flow sensor in this mode, so RC cannot detect real flow — it can only assume a section switched off stops that section's share of the material. Fewer sections than dispensers makes applied totals read high; more sections than dispensers makes them read low (and the machine genuinely over-applies, since the dispenser keeps running and simply overlaps). A reminder appears each time the Manual Rate value is entered. A single-dispenser machine needs nothing special.

Controls

Control	Action
Navigate left	Previous product
Navigate right	Next product
Cancel	Discard changes
Save	Save settings

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8. Products – Data Reset

Menu: Products > Data

Displays accumulated totals for the selected product and allows resetting individual counters. Navigate between products with the left/right arrows.

Counter	Description
Coverage Area 1	Total area covered (primary counter)
Coverage Area 2	Total area covered (secondary counter)
Quantity Applied 1	Total product applied (primary counter)
Quantity Applied 2	Total product applied (secondary counter)
Hours 1	Operating hours (primary counter)
Hours 2	Operating hours (secondary counter)

Press the reset button next to any counter to zero it. Both counters are independent, allowing a session total and a season total to be tracked simultaneously.

Controls

Control	Action
Navigate left	Previous product
Navigate right	Next product
Reset counter	Zero the selected counter
Cancel	Discard changes
Save	Save settings

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9. Products – Monitoring

Menu: Products > Monitoring

Displays real-time diagnostic data for the selected product. Navigate between products with the left/right arrows.

Rate Information

Display	Description
Target UPM	The current setpoint (desired flow rate)
Applied UPM	The measured actual flow rate
Rate Error %	Difference between target and actual, with sign
PWM	Current PWM output value (0–255)
Frequency (Hz)	Flow sensor pulse frequency

Speed & Coverage Information

Display	Description
Ground Speed	Current speed in MPH or KPH
Working Width	Implement working width in feet or meters
Work Rate	Area covered per hour (ac/hr or ha/hr)

Section Status

Shows 16 section indicator boxes — green for On, red for Off.

Module Communication

Display	Description
WiFi / Signal	Module signal strength (RSSI) bar
Counts per Revolution	The number of pulses the sensor produces per revolution. Total pulses per minute are divided by this value to calculate RPM. For example, a value of 3 means every 3 pulses equals one revolution.

Controls

Control	Action
Navigate left	Previous product
Navigate right	Next product
Cancel	Discard changes
Save	Save settings

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10. Products – Control (PID Tuning)

Menu: Products > Control

Configures the PID controller that adjusts the motor or valve to maintain the target rate.

Primary Settings

Setting	Description
1. Gain	Rate adjustment factor — how strongly the controller responds to rate errors. The effect is linear: doubling Gain doubles the correction. Errors are measured as a percentage of the target rate, so the same Gain setting behaves consistently at low and high rates.
2. Integral	Trims residual error once the rate is close to target (inside the BrakePoint band — see below). Far from target, only Gain (proportional) drives the approach; Integral does not speed that up. Increase if the rate settles near, but not exactly on, target. Integral accumulates once per PID loop tick, so a shorter PID loop time (setting 4 below) also makes it trim faster in real time, independent of this setting.
3. Max Power	Maximum PWM power delivered to motor or valve
4. Min Power	Minimum PWM power delivered to motor or valve

Adjustment Process

1. **Initial Setup** — Set minimum power to the minimum needed to move the motor/valve. Set maximum power to 100. Set Integral to 0.
2. **Gain Adjustment** — Slowly increase gain until the system overshoots the target, then reduce slightly to stabilize.
3. **Integral** — If the rate approaches target but settles slightly off, increase Integral to close the remaining gap. It only acts once you're already close (see BrakePoint) — it will not speed up a slow ramp toward target.

Optional / Advanced Settings

Settings 7–9 apply to the Combo Timed control type only; setting 5 applies to Motor only.

Setting	Description
1. Deadband	Error % below which no adjustment is made. Also controls when accumulated Integral is reset — only once the error clearly crosses to the other side of target, not on small wobble near the setpoint.
2. BrakePoint	Error % (of peak target) defining the "near target" band. Outside this band, only fast proportional (Gain) control acts and Integral stays at zero. Inside it, proportional gain drops to the Slow Adjustment % and Integral begins accumulating.
3. Slow Adjustment	% of full proportional adjustment used once inside the BrakePoint band
4. PID loop time	How often (50–250 ms) the PID recalculates its output. A shorter time makes the valve/motor react sooner to a real rate change, and also speeds up Integral trimming in real time (see Integral, above) since it accumulates once per tick. Reducing this after Gain/Integral were tuned at a slower rate can introduce new oscillation — check for overshoot as with any Gain increase.
5. Slew Rate	Motor control type only — maximum total PWM change per PID loop. Has no effect on Standard/Combo Close valves, which are positioned fresh each loop rather than ramped.
6. Max Integral change	Maximum Integral PWM change per PID loop. The total accumulated Integral is separately capped to the Max Power − Min Power range, so it can reach full authority (e.g. to free a stuck valve) without jumping there in one loop.
7. Min Start %	Combo Timed control type only — % of target rate below which the adjust/pause cycle is skipped for a faster start from 0.
8. Adjust time	Combo Timed control type only — milliseconds the valve is actively adjusted during each cycle.
9. Pause time	Combo Timed control type only — milliseconds the valve pauses (holds) between adjustments.
10. Min flow sensor Hz	Pulse frequency below which the sensor is treated as stopped/no-flow
11. Max flow sensor Hz	Pulse frequency above which pulses are rejected as sensor noise/bounce
12. Flow sample time (Smp Tm)	Time window in milliseconds (50–1500) used to calculate the median flow reading. A longer window smooths the reading; a shorter window responds faster. Default 400 ms.

Controls

Control	Action
Navigate left	Previous product
Navigate right	Next product
Load defaults	Restore default PID values
Open rate graph	Display the rate graph
Gain/Integral/Power +	Increase the selected value
Gain/Integral/Power -	Decrease the selected value
Show advanced settings	Expand the advanced/optional settings panel
Cancel	Discard changes
Save	Save settings

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11. Machine – Calibrate

Menu: Machine > Calibrate

Use this screen to determine the **Cal Factor** (pulses per unit) for a product. The goal is to achieve a consistent meter roller RPM or liquid flow rate matching typical field operation.

Step-by-Step Calibration

① Step 1 — Set Ground Speed

- Enter the ground speed you typically operate at.

② Step 2 — Set Meter Roller RPM / Valve Position

- Switch the product **ON** using the power button.
- Enter an **Initial Base Rate** and an **Estimated Cal Factor**. The Base Rate becomes the product's working rate when Save is pressed.
- Press **Start** — RC adjusts flow to match the target rate.
- If successful, RPM is locked.
- The PWM value used is shown in the **PWM** box (for motor controllers). While locked and stopped it can be edited — accept the value RC found, or enter your own (0–255).
- RC stops automatically when the target rate is met.

If RC fails to reach target RPM:

- Press **Stop**.
- Check ground speed.
- Adjust the estimated Cal Factor, PWM min/max, or metering drive range.

③ Step 3 — Set Actual Cal Factor

- Press **Start** to run the meter roller and collect a sample.
- Press **Stop**, then enter the measured amount collected.
- RC calculates and displays the new Cal Factor.
- Press **Save** to record it.

④ Step 4 — Optional Adjustments

- To redo meter speed: press **Unlock**, then repeat Step 2.
- To reuse a previously saved setting: press **Lock**, then proceed to Step 3.
- To skip Step 2 entirely (motor controllers): press **Lock**, enter the desired **PWM** directly, then proceed to Step 3. Useful for iterating on the metering-unit opening at a fixed PWM — run, weigh, adjust the opening, repeat.
- The lock is saved with Save and remembered the next time the product is calibrated, so Step 3 can be repeated without redoing Step 2.

Calibrating from a Phone

The **Calibration** page on the phone runs the same calibration as this screen, so Start, Stop and the measured amount can be entered standing at the meter or the catch container. To connect a phone, see Options – Phone Tab.

- The page has the product buttons 1–5 (lit when that product's power is on), **Power**, the lock (shown as **Meter set** / **Meter not set**), **Start** / **Stop**, the live Pulses and Hz, and the Speed, Base Rate, Meter Cal, Expected, Measured and PWM boxes, with **Save** / **Cancel**.
- Tap a box to enter a value. The old value is shown faded; tapping away without typing keeps it.
- The tablet and the phone show the same calibration, and either can start or stop it. The tablet screen follows a run started from the phone.
- **A run started from the phone stops by itself** if the phone loses its connection, or its screen is locked or switched to another app, for about 5 seconds. The page then shows *Last run stopped: phone connection lost*.
- **A run started on the tablet does not stop by itself.** Start from the phone if you will be away from the tablet. Stop works from either.
- Closing the tablet's Calibrate screen stops a run started on the tablet; a run started from the phone keeps going.

Important Notes

- AgOpenGPS is **not** used during calibration.
- The Switchbox is **not** used during calibration.
- If a switchbox is present, ensure the **master switch is OFF** before calibrating.
- Calibration uses **Constant UPM, Mode 2**.
- All sections on the calibrating module are activated during calibration; modules that are not calibrating are unaffected.
- More than one product can be calibrated at the same time, on the same or different modules (requires up-to-date module firmware).
- After calibration, the original mode and section states are restored.
- If RateController closes during a calibration (power loss, a crash, or ended in Task Manager), the product's original settings and the speed source are restored at the next start, with the message *Product N restored after an interrupted calibration*. A calibration that wasn't saved has to be repeated.
- Start is refused if a locked motor product has PWM 0 — enter a PWM value first, or the motor would not turn.
- Watch the **Hz** display while running: aim for at least 10–12 Hz at working rate for a stable reading. Very high frequencies can exceed the sensor's switching limit — if Hz plateaus while speed rises, the sensor is at its ceiling.

Controls

Control	Action
Start	Start the meter roller / begin collecting a sample
Stop	Stop the meter roller
Lock/Unlock	Lock or unlock the current RPM setting. Shown on the phone as Meter set / Meter not set. Saved with Save
Power on/off	Switch the product on or off
Cancel	Discard changes
Save	Save the calculated Cal Factor
PWM	PWM used while locked (motor controllers only). Editable when locked and stopped; press Save to keep the value
Hz	Live sensor pulse frequency for the selected product (shown for all control types)

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12. Machine – Sections

Menu: Machine > Sections

Defines the boom sections and their widths.

- There are **16 sections per module** and up to **8 modules**.
- Module 0 = sections 1–16
- Module 1 = sections 17–32, etc.
- Sections can be defined **individually** or grouped into **zones**.
- All sections in a zone share the same width.
- All sections in a zone are controlled by one switch.
- AgOpenGPS can still auto-control each section individually within a zone.
- When creating a new profile, sections must be defined.
- Sections are also automatically redefined when AgOpenGPS section count changes.

Control	Action
Copy Button	Copies the width from Section 1 to all other sections
Cancel	Discard changes
Save	Save settings

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13. Machine – Relays

Menu: Machine > Relays

Assigns relay functions to each relay output on the module. There are **16 relays per module** and up to **8 modules**.

Control	Action
Reset (circle)	Reset all relays to default
Renumber (+)	Renumber sections sequentially
Relay Number	The relay output number
Relay Type	The behavior of the relay (see below)
Number	For Switch type: switch number. For Section/Invert_Section type: section number.
Cancel	Discard changes
Save	Save settings

Application Control

Spraying occurs when:

- Master is ON
- A section is enabled
- In Auto, the guidance system allows application

Relay Types

Type	Description
Section	Controls an individual section. In Manual, the switch turns the section on or off. In Auto, the guidance system controls the section, but the switch can disable it.
FlowMaster	Controls overall product flow. In Manual, it follows the Master switch. In Auto, it is on only when the Master switch is on and the machine is moving. Use this to control overall product flow through an upstream valve or common shutoff.
Invert_FlowMaster	Opposite of the FlowMaster output — on when the master switch is off or (in Auto) the machine is stopped. Intended for valve wiring that requires an inverted signal, not as a bypass for section activity. Use **Bypass** if you need a relay that responds to whether sections are actually on.
Bypass	On when all sections are off; off when any section is on. Use this for a return or bypass valve that diverts product when no boom sections are applying.
Master	Follows the Master switch. Use this when a relay should follow the master logic itself rather than actual flow-enabled status.
Invert_Master	Opposite of the Master output (on when the Master output is off).
Slave	On when any section is on; off when all sections are off.
Invert_Section	Opposite of the Section output (on when the section is off).
Power	Always on.
Switch	Controlled directly by a switch (not affected by Master or Auto).
Hyd Up	Hydraulic raise.
Hyd Down	Hydraulic lower.
Tram Right	Tram marker right.
Tram Left	Tram marker left.
Geo Stop	Geofence stop output.
None	No function assigned to the relay.

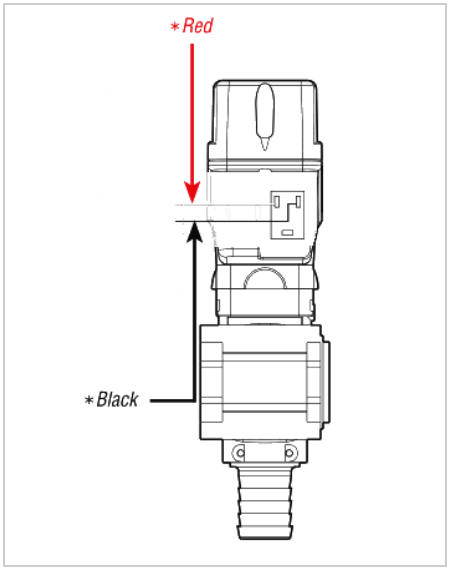
FlowMaster Valve Options

These options apply only when exactly one **FlowMaster** relay is defined on the selected module.

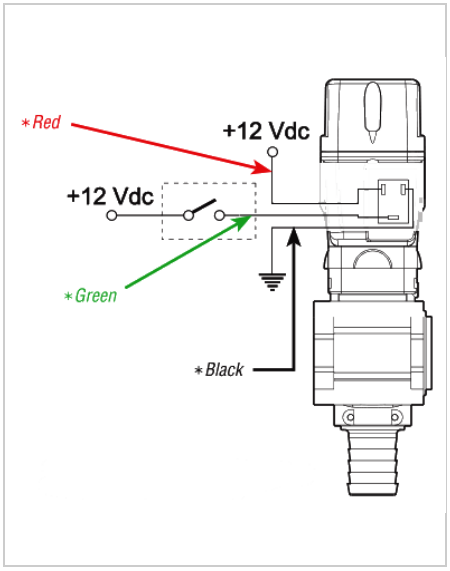
Option	Description
2-wire valve (single relay powers open and close)	One **FlowMaster** relay drives both open and close directions using the module's H-bridge output. Requires module support (e.g. RC15).
2-wire valve (2 relays: FlowMaster powers open, Invert_FlowMaster powers close)	Two relay outputs: the **FlowMaster** relay supplies power to open the valve; the **Invert_FlowMaster** relay supplies power to close it. Use this on modules without H-bridge support (e.g. RC11-2).
3-wire valve (relay signal, valve self-powered)	One **FlowMaster** relay provides a signal. The valve uses its own power supply to open or close based on that signal.

Valve Wiring Reference

The diagrams below show the wiring difference between 2-wire and 3-wire valves.



2-wire valve



3-wire valve

↑ Contents

14. Machine – Switches

Menu: Machine > Switches

Configures how the master switch, auto switches, and on-screen virtual switches behave.

Master Switch Mode

Select one option. Controls what the master switch turns on and off.

Option	Description
Standard	Master switch controls the master relay and all section relays. When master turns off, all section relays turn off. When master turns back on, section relays resume their switch positions.
Master Override	Forces the module to treat master as always on, regardless of switch state. Useful for motor control when the master switch should not interrupt the motor PID loop.

Switch Type

Select one option. Describes the physical type of master switch connected.

Option	Description
Momentary	Standard push-to-make switch. Each press toggles master on or off. Holding in the On position activates Primed Start after the configured delay.
Maintained	Toggle or latching switch. Master follows the switch position directly — on when switch is on, off when switch is off. Primed Start is not available in this mode.

Other Settings

Setting	Description
Work Switch	Uses a switch connected to the implement to gate master on/off. The switch's pin, momentary behavior and NO/NC invert are set per module under Modules > Pins.
Auto Switch – Rate	Enables automatic rate control via the switchbox Auto Rate switch
Auto Switch – Sections	Enables automatic section control via the switchbox Auto Section switch
On-Screen Switches – Enabled	Shows the virtual on-screen switch panel (see below)
On-Screen Switches – Rate Only	Shrinks the panel to just the rate up/down buttons. Master, PRM, the section switches and Auto Rate/Section are hidden, and the switch count is not used.
On-Screen Switches – 4 / 8 / 12 / 16	Number of section switches shown on the on-screen switch panel. Not available when Rate Only is selected.

On-Screen Switch Panel

When enabled, a floating switch panel appears on screen. Its buttons always show the switch states in use. With no physical switchbox connected, the panel is the switchbox. With a switchbox connected the panel still works — see With a Switchbox Connected below.

Button	Function
MST	Toggles master on/off. Green = on, unlit = off.
PRM	Activates Primed Start immediately. Green while priming is active. Dimmed when Maintained switch type is selected or when the vehicle is moving.
▲ / ▼	Rate adjustment. Hold to increase or decrease rate or PWM.
1 – 16	Section switches (4, 8, 12, or 16 buttons, as configured). Green = on, unlit = off.
Auto Rate	Toggles automatic rate control.
Auto Section	Toggles automatic section control.

The panel has no title bar and uses the display colours set under Options. To reposition it, drag the panel background (the gaps between the buttons) with either mouse button, or **right-click and drag** on a button. A left click on a button always presses it. Where it is dropped decides what it follows. Dropped overlapping the main screen it follows the main screen. Dropped overlapping another floating window — a product popup display or the pressure display — it follows that window instead, so a group of windows can be moved together. If it overlaps both, the main screen wins. Drop it clear of everything to unlatch it. With **Rate Only** checked the panel shrinks to a small window holding just the ▲ / ▼ buttons.

> **Note:** Before pressing PRM, turn on the section switches you want to prime. Only sections that are currently switched on will run during priming. Partial priming (some sections only) is valid.

With a Switchbox Connected

- **Taking control:** the first on-screen press takes control from the switchbox. It starts from the current switch positions, so only the button pressed changes.
- **The switchbox always wins:** moving any switch on the switchbox (master, sections, rate or auto) hands control back straight away and the switchbox positions apply again. The work switch doesn't count.
- **No timeout:** the screen keeps control until a switchbox switch is moved. Unlike the phone, there is no 120 second timeout.
- **▲ / ▼:** change the rate while held and stop when released.
- **Main screen MST and Auto:** work the same way. With no switchbox, MST on from the main screen also turns all enabled sections on. With a switchbox connected it only turns master on — the sections stay as they are.

Plugging In or Unplugging the Switchbox

Neither changes any switch positions. About 3 seconds after the switchbox stops communicating, the on-screen panel carries on from the positions in use, including a maintained master that is on. When the switchbox is plugged back in, those positions are held until a switch on the switchbox is moved, and then the switchbox takes over. The exception is the first switchbox after the app starts: its switch positions apply straight away.

Phone Switches

The **Switches** page on the phone is a copy of the on-screen switch panel: MST, PRM, ▲, ▼, the section switches, Auto Rate and Auto Section, in the tablet's display colours. It shows the full panel even when the tablet uses Rate Only. Like the on-screen panel, it works with a physical switchbox connected. To connect a phone, see Options – Phone Tab.

- **Taking control:** the first press on the phone takes control from the cab. It starts from the current switch positions, so nothing changes until a button is pressed. The status line shows *Remote in control* and the seconds left.
- **The cab always wins:** moving any switch on the switchbox (master, sections, rate or auto) hands control back straight away and the switch positions apply again. The work switch doesn't count. With no switchbox connected, a press on the on-screen panel does the same. The phone shows *Cab switchbox took control*.
- **Sharing with the on-screen panel:** with a switchbox connected, on-screen presses don't end the phone's control — the phone and the screen share it until a switchbox switch is moved. While only the screen has control, the phone shows *Cab in control*.
- **Timeout:** after 120 seconds with no press on the phone or the on-screen panel, control goes back to the cab and **master turns off** (*Timed out, master off*). A maintained master switch stays off until it is switched off and on again.
- **Auto Section:** can be turned off from the phone, for when auto was left on in the cab. While it is on and AgOpenGPS is connected, AgOpenGPS controls the sections and the page shows a warning.
- **▲ / ▼:** change the rate while held. If the phone loses its connection while a button is held, the change stops within about a second.
- **PRM:** works as on the tablet: master on and prime, only with a momentary master switch and the machine stopped. When priming ends without Resume After Prime, master turns off and the sections stay as set on the phone.

Controls

Control	Action
⊗ Cancel	Discard changes
Save	Save settings

↑ Contents

15. Machine – Pressure

Menu: Machine > Pressure

Calibrates a pressure sensor using two known reference points, and sets a maximum pressure limit per module.

Calibration Procedure

1. Select the **Module** the pressure sensor is connected to.
2. Set the system to a **known low pressure**.
3. Enter the sensor **reading** and the actual **pressure** value.
4. Set the system to a **known high pressure**.
5. Enter the sensor **reading** and the actual **pressure** value.
6. Enter the **minimum sensor reading** that corresponds to 0 pressure.
7. Press **Show** to display the live pressure screen.

Pressure Display Window

Show opens a floating window with the live pressure for the current product's module. It has no title bar and uses the display colours set under Options.

Click and drag the window (either mouse button) to reposition it. Where it is dropped decides what it follows. Dropped overlapping the main screen it follows the main screen. Dropped overlapping another floating window — a product popup display or the on-screen switch panel — it follows that window instead, so a group of windows can be moved together. If it overlaps both, the main screen wins. Dropped clear of everything it follows nothing until it is dropped onto something again.

The window detaches while the main screen is minimized and re-attaches when the main screen is restored. If the window it was following is closed, it stays where it is and simply stops following. Its position is remembered between sessions.

Max Pressure

Sets the maximum allowed pressure for the selected module. When the measured pressure exceeds this value:

- The application shows a **High Pressure** alarm.
- The limit is also sent to the module, which reduces flow on its own to bring the pressure back below the limit — this protects the system even if communication with the PC is lost. Requires up-to-date module firmware.

If the pressure limit trips repeatedly in a short time, the module treats it as a fault and keeps reducing flow until the **master switch is turned off**, which resets the protection.

Set Max Pressure to **0** to disable the limit.

Controls

Control	Action
Cancel	Discard changes
Save	Save settings

↑ Contents

16. Machine – Primed Start

Menu: Machine > Primed Start

Allows boom priming or a simulated product application while the tractor is stopped. The tractor must not be moving for primed start to activate.

Setting	Description
On Time	How long the application runs (seconds)
Speed	Simulated ground speed used to calculate application rate during priming
Master Switch Delay	How long to hold the physical master switch in the On position to trigger primed start (seconds). Applies to momentary switchbox only.
Resume	After On Time expires, if the tractor is moving, continue with normal operation

Activating Primed Start

Physical Switchbox — Momentary Switch

Hold the master switch in the **On** position for the duration of the **Master Switch Delay**. Primed start activates automatically once the delay expires.

On-Screen Switch Panel

Press the **PRM** button. Primed start activates immediately — no hold required. The PRM button turns green while priming is active.

Before pressing PRM, turn on the section switches you want to prime. Only sections that are currently switched on will run during priming.

Cancelling

Press the master switch Off, or press the on-screen **MST** button while master is on. Primed start stops immediately.

> **Note:** Primed Start is not available when **Maintained** switch type is selected in Machine > Switches.

Controls

Control	Action
⊗ Cancel	Discard changes
Save	Save settings

↑ Contents

17. Modules – Boards

Menu: Modules > Boards

Lists every board RateController can hear, and is where a board is given its module number and description. Each board is known by its built-in hardware (MAC) number, so boards are told apart even when they are the same type or share a module number. The list updates every second.

Board List

Column	Description
Module	The board's module number, 0–7, or **New** for a board that has no module yet.
Board	The board type, the last part of its MAC number, and its description, e.g. *ESP Rate ...0B:B0:12 Left Tank*.
State	**Connected** — the board is reporting. **Duplicate** — two or more boards are on the same module. Each is listed on its own row. Move one of them to a free module. **Update firmware** — a board on this module has firmware too old for this version of RateController. The module is not used until the firmware is updated. **New** — a new or blank board waiting for a module. New boards are listed last. **→ 3, not sent** or **Not sent** — a saved change that goes out with the next Send.

Settings

Select a board in the list to change it.

Setting	Description
Module	The module the board serves. The list holds the board's own module and the free modules only: a module that already has a board, including one that needs a firmware update, is not offered.
Description	A name for the board, up to 16 characters, e.g. "Left Tank". It is stored on the board and kept through firmware updates. The box can't be used for a Nano (it stores no description), for a board on a duplicate module, or for a new board until a module is chosen.
Web page link	For an ESP32 on WiFi, the address of its web page. Click it to open the page. An ESP32 on Ethernet shows *Ethernet, no web page*.

Setting Up a New Board

New and blank boards start without a module (ID 25). While one is reporting, the **Boards** button on the menu has an orange border.

1. Connect the board to the same network as RateController. An ESP32 on WiFi is set up from Modules > Communication > WiFi instead, which gives it its module at the same time.
2. Open **Modules > Boards** and select the board marked **New**.
3. Choose a free **Module**, add a **Description** if you like, and press **Save**. The module pages switch to the chosen module.
4. RateController offers the starting configuration for the board's type (Nano, Teensy or ESP32). Press **OK** to load it, then check the Config, Pins, Relay Pins and Valves pages against your own board.

5. Press **Send** ⬆ . The board takes its module number and settings, restarts, and reconnects as that module.

Moving a Board or Changing its Description

1. Select the board, choose the new **Module** and/or type a new **Description**, and press **Save**. The State column shows the change as *not sent*. A module change also moves the module pages to the new module, whose settings will be sent to the board.
2. Press **Send**. The board restarts on its new module with that module's settings.

- Only one module change can wait for Send at a time. Descriptions can be saved for any number of boards.
- To undo a saved module change, select the board's own module again and press Save. The module pages go back to the board's module.
- Another board can't be selected while the current one has unsaved edits. Press Save or Cancel first.

When Send is Refused

Send does not go out, and a message says why, when:

- the board being moved has stopped reporting, or another board has appeared on its new module;
- the module being edited has no board connected, or its board needs a firmware update;
- a module page has unsaved changes after a module change ("*Save or cancel the pending changes first.*"). Save or cancel them and press Send again.

Except for the last case, saved module changes and descriptions are dropped when Send is refused, so they can't go out later by surprise. Save them again when the board is ready.

> **Note:** A Nano has no description. It is shown by its MAC number.

> **Note:** A board that is switched off isn't seen, so its module looks free. Make sure every board is powered before moving boards around.

> **Note:** A board with older firmware may not appear here at all. An older Nano, for example, isn't listed and its module shows as not connected. Update the board's firmware, then it appears as its module or as New. See Updating Module Firmware (§30).

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18. Modules – Communication

Menu: Modules > Communication

Configures how the application communicates with rate control modules. There are three tabs: **Ethernet**, **WiFi** and **CanBus**.

Modules talk to the application either over the network (Ethernet, or WiFi for ESP32 modules) or over CAN Bus. Tick **Enabled** on the Ethernet tab or on the CanBus tab and press Save; ticking one unticks the other. Ethernet is the default. With CAN Bus, Ethernet stays active for AgOpenGPS.

Ethernet

Chooses the wired network adapter the modules are plugged into, and moves the modules to that adapter's subnet.

Control	Description
Enabled	Use the network for module communication (unticks CAN Bus). Applied on Save.
Adapter	This PC's wired adapters that have an address, shown as name and subnet, e.g. *Ethernet 2 (192.168.1.x)*. WiFi and the tablet hotspot are not listed. If the saved subnet is on no adapter of this PC, such as after a new install or an adapter change, the page picks one by itself: the adapter modules are heard on, otherwise the first.
Status line	*Modules on this adapter: N*, *No modules heard on this adapter. Press Send to put them on it.*, or *No wired adapter on …* when the cable is unplugged.
Refresh	Reads the adapter list again, after plugging in a cable or an adapter.
Send	Tells the modules to move to the selected adapter's subnet. They restart and are counted on the status line once heard. After choosing a different adapter, Send stays greyed out until Save is pressed.

The steps are shown on the page: select the adapter, Save, then Send.

> **Note:** An ESP32 connected over WiFi ignores Send, so a subnet meant for the cable doesn't take it off its WiFi network.

WiFi

Sets up an ESP32 module to join a WiFi network from here, instead of from the module's own web page. Setup also gives the module its ID. The module can join either the tablet's own hotspot or the WiFi network this PC is connected to.

Control	Description
Tablet Hotspot	The module joins this tablet's Windows Mobile Hotspot. RateController keeps the hotspot on while this is chosen. It is applied on Save, and the hotspot is started every time RateController starts. The hotspot is set to 2.4 GHz, because the module can't use 5 GHz. See Phone Tab for more on the tablet hotspot.
Current WiFi Network	The module joins the network this PC is connected to in Windows, such as a shop router or a phone hotspot. Connect the PC to it first.
Network	The network the module will join. For the hotspot: its name and state, which is On, Off, Checking..., *firewall not allowed*, *Off, restarts failed*, or Windows' reason when it can't start. For the PC's network: its name, or *Not connected*.
Hint link	For the hotspot: *If necessary, turn off Power saving in Hotspot settings*. Windows turns the hotspot off after 5 minutes with no device connected. For the PC's network: a link to **WiFi settings**.
Band line	The hotspot's band. It also warns when this PC's WiFi is on 5 GHz: the module needs the network's 2.4 GHz band.
Discovered modules	Module hotspots in range, with signal strength. A module's hotspot is named **RateModule*_ID_6 characters***, e.g. *RateModule_25_1A2B3C*; a new board shows ID 25. Select the module to set up. A lone module is selected by itself. Selecting a module that already has an ID picks that ID, and selecting a new board picks the lowest free ID. The refresh button scans again.
Network password	The password of the PC's network. It is remembered for that network. It isn't needed for the tablet hotspot, whose password the application knows.
Module Password	The module's own hotspot password. Leave it blank when the module hotspot is open (the default). It is remembered.
Selected Module ID	The ID the module gets, 0–7. Connected IDs are marked *(Connected)*. The default is the lowest free ID. Picking an ID selects that module's hotspot in the list, if it's there.
Address link	The web page address of the selected module, when it's a connected ESP32 on WiFi. Click it to open the page. Other boards show *No web page*, and an ESP32 on Ethernet shows *Ethernet, no web page*.
Set Up	Starts the setup (see below).
Cancel	Stops a setup. The PC is put back on its own network first.
Log	Each step of the setup is written in the box. The log button opens the whole log in a larger window.

Setting Up a Module

1. Choose **Tablet Hotspot** or **Current WiFi Network**, and press Save if it changed.
2. Power the module and wait for its hotspot to appear in the list. Select it.
3. Check the **Selected Module ID**, and type the network password if needed.
4. Press **Set Up**. The PC leaves its network for a moment, joins the module's hotspot and sends it the network name, password and ID. The module restarts and joins the network. The PC goes back to its network and waits up to a minute for data from the module, then asks the module what went wrong if nothing arrives. The log ends with *Done: module N is connected.* when it worked.

The page asks first if the chosen ID is already connected (only choose it when setting up that same module again), or if the PC's network password is blank (only for an open network). After a successful setup the list is scanned again, and the next free ID is chosen for the next module.

Firewall

Windows Firewall can block module data on a phone hotspot or router network. On the first visit to this tab in a session, and after a setup where no data arrived, the page asks to allow it. Windows then asks for administrator permission. The tablet hotspot needs its own permission the first time it is saved. If that's declined, the Network line shows *firewall not allowed*.

> **Note:** The ESP32's radio is 2.4 GHz only. A network that is 5 GHz only can't be used. A dual-band router works when its 2.4 GHz band has the same name.

Module WiFi (ESP32)

ESP32 modules run their own hotspot and settings page. The WiFi tab above does the usual setup for you. The page is for checking the module or changing it by hand. Connect a phone, tablet or laptop to the module's hotspot, named **RateModule_*ID*_** followed by six characters unique to that board. Then browse to **192.168.*(200 + module ID)*.1**: module 0 is at 192.168.200.1, module 1 at 192.168.201.1, and a new board (ID 25) at 192.168.225.1. Press **Network**. The hotspot is open unless a hotspot password has been set.

The Network page is where the module is pointed at your own WiFi network, so it can reach the application over an existing network instead of you connecting to the module directly.

Control	Description
Network	Name of the WiFi network the module joins
Password	Password for that network
Use this Network	Ticked, the module joins the network above. Unticked, the module runs its hotspot only.
Scan for Networks	Lists the networks in range with signal strength, channel, and a padlock on encrypted ones. Tap a network to fill in its name.
Retry Connection Now	Forces an immediate connection attempt. Shown only when the module is set to join a network and is not connected.
Hotspot Password	8 to 10 characters. Leave empty for an open hotspot.
Save/Restart	Saves the settings and restarts the module

> **Note:** Tap a network in the scan results *before* typing the password. Tapping reloads the page, which clears anything already typed.

The status line below the settings reports what the module is doing:

Status	Meaning
Wifi Connected to *name* (address) on channel *n*	Joined. The address shown is the module's address on your network.
Wifi Not Connected — retrying in the background	Set to join a network but not connected. The module keeps trying.
Password refused by *name*	The network rejected the password. Correct it and press Save/Restart.
Hotspot only	Use this Network is unticked.

Joining and Retrying

A module set to join a network keeps trying for as long as the setting is ticked. It does not switch itself back to hotspot-only, and it does not restart itself, so a network that goes down — the shed WiFi switched off, or the machine driving out of range — no longer interrupts work in progress. Earlier firmware gave up after several failures, turned **Use this Network** off and rebooted the module.

Attempts are deliberately spaced out. The module has a single radio shared between its hotspot and its connection to your network, so every attempt takes the hotspot briefly off the air. While a device is connected to the hotspot — a tablet running the application, for instance — attempts are about two minutes apart, or about five minutes if the module has never successfully joined that network. With nothing on the hotspot the module tries more often. Press **Retry Connection Now** rather than waiting.

Loading the module's settings pages pauses the retries for a minute after the last page, so the pages stay responsive while you are configuring. **Retry Connection Now** works regardless.

Scanning for networks takes the radio off the hotspot for a second or two, so a brief pause after pressing **Scan for Networks** is expected.

> **Note:** The hotspot uses the same channel as the network the module joins. Devices on the hotspot drop once the first time a module joins a network, and again if that network later changes channel — they reconnect on their own. From then on the module starts its hotspot on the remembered channel.

Some routers refuse a wrong password in a way the module cannot tell apart from a weak signal. When that happens the status stays on **Wifi Not Connected** instead of naming the password, so check the password whenever a module will not join a network you know is in range.

CAN Adapter

Driver	Description
SLCAN	Serial port-based CAN adapter. Requires COM port selection.
InnoMaker USB2CAN	Native USB CAN adapter (InnoMaker hardware). No COM port required.
PCAN	Native USB CAN adapter (Peak PCAN hardware). No COM port required.

COM Port (SLCAN only)

1. Use the dropdown to select the COM port the SLCAN adapter is connected to.
2. Press **Refresh** to rescan available COM ports.

Diagnostics

Option	Description
Diagnostics	Enables CAN traffic debug logging in the activity log

Status Indicators

Indicator	Meaning
Module	Green — module data received within the last 2 seconds
Driver	Green — CAN adapter is open and receiving frames within the last 4 seconds

> **Note:** Changing the CAN adapter type or COM port while CAN Bus is active will restart the CAN bridge with the new settings.

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19. Modules – Config

Menu: Modules > Config

Configures the hardware settings of the connected module.

Module Identity

The underlined line at the top shows which module the module pages are editing — the module number, the board's description (or reported label), and its connection state. It updates live while the page is open. If it turns red with **"Two boards are answering as module N!"**, two powered boards share the same module ID — disconnect one and give it its own ID before continuing.

If the module's board needs a firmware update, the line turns red and reads **Module N (Firmware update required)**. The module is not used, and nothing is sent to it, until its firmware is updated.

Below it, while the module is connected, is the board type it reports, e.g. **Teensy Rate** or **ESP Rate**. Nothing is shown while it is not connected.

Boards are given their module number and description on the Modules > Boards page (§17), which also lists new boards.

Setting	Description
Module to edit	The module to edit, listed as *N (Connected)*, *N (Not connected)* or *N (Update firmware)*. All module pages (Config, Communication, Pins, Relay Pins, Valves) and the Send button work on this module; other connected modules ignore the settings. A new choice takes effect straight away, without Save. It is refused while any module page has unsaved changes.

Load Preset

Loads a full starting configuration for a known board — the settings on this page, the sensor pins and the relay pins — and saves it. It uses the type of the board reporting on this module. When no board is reporting, it asks which type to load.

Preset	Microcontroller
Nano (RC12-3)	Arduino Nano
Teensy (RC11-2)	Teensy 4.1
ESP32 (RC15)	ESP32

After loading, check the Pins, Relay Pins and Valves pages against your own board, then press **Send**. Load preset is refused while any module page has unsaved changes. **Rescan** resets the settings on this page only, to be saved or cancelled like any change.

> **Note:** A preset replaces the module's pin and relay settings. Only use it during initial setup or when changing hardware. It does not change the module number.

Sending to the Module

Press **Send** to give the module its settings. Send is refused, with a message, when no board is connected on the module being edited, or when its board needs a firmware update. Saved but unsent changes on the Boards page are dropped when that happens; save them again when the board is ready.

1. Relay Type

Value	Description
0	No relays
1	GPIOs — use micro-controller pins directly
2	PCA9555 — 8-relay module
3	PCA9555 — 16-relay module
4	MCP23017
5	PCA9685
6	PCF8574

2. Pin Assignments by Board

Parameter	RC5 (Nano)	RC8 (Nano)	RC11 (Teensy)	RC11-2 (Teensy)	RC12 (Nano)	RC12-3 (Nano)	RC15 (ESP32)
Sensor Count	2	2	2	2	1	2	2
Relay Control	4	4	1	1	2	4	5
Flow 1	2	2	28	28	D3	3	17
Flow 2	3	3	29	29	—	2	16
DIR 1	4	4	37	37	D6	4	32
DIR 2	6	6	14	14	—	6	25
PWM 1	5	5	36	36	D9	5	33
PWM 2	9	9	15	15	—	9	26
Work Pin	—	—	—	30	—	15	—
Pressure Pin	—	—	—	40	—	14	—

3. Nano Pin Reference

Pin Name	Number
D0–D13	0–13
A0–A7	14–21

Controls

Control	Action
Cancel	Discard changes
Save	Save settings

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20. Modules – Pins

Menu: Modules > Pins

Manually assigns the GPIO pins used for sensors and motor/valve control on the module. The grid shows one row per sensor (the sensor count is set in Modules > Config). Pin values range from 0–50. A blank value (displayed as "–") means the pin is not assigned.

Grid Column	Description
Sensor	Sensor/product number (read-only)
Flow	Input pin for the flow sensor
Dir	Output pin for the motor/valve direction channel
PWM	PWM output pin for the motor/valve
Bin	Input pin for the bin-empty sensor. Leave unassigned ("–") if the product has no bin sensor — the bin alarm is off for that product.
Invert	Inverts the bin sensor reading. Unchecked: an open input reads *empty* (sensor closes to ground while material is present). Checked: a grounded input reads *empty*.

Setting	Description
Work Pin	Input pin for the work/implement switch
Momentary	Enables momentary switch behavior on the work pin
Invert Work Switch	Inverts the work switch reading to suit the sensor type. Unchecked: an open input reads as *working* (normally-closed sensor). Checked: a grounded input reads as *working* (normally-open sensor). In momentary mode the toggle occurs on the opposite edge. Older module firmware ignores this setting.
Pressure Pin	Analog input pin for the pressure sensor

Press **Rescan** to clear all pin assignments. Duplicate pin numbers are rejected on Save.

> **Tip: Load preset** in Modules > Config fills in these pin values for a standard board. Only change them if your hardware differs from the standard layout.

Bin-Empty Alarm

When a Bin pin is assigned, the module watches that input and reports the bin state to RateController. The bin sensor input uses an internal pull-up, so a simple switch or level sensor wired between the pin and ground is all that is required. The module filters the reading — the sensor must hold its state for about 2.5 seconds before the reported state changes, so brief flickers from material sloshing are ignored.

- The alarm is only raised **while the master switch is on** — an empty bin at start-up does not alarm until you begin applying.
- When it triggers, the alarm button shows **Bin Empty** with the product name, the product's tile turns red, and on the selected product the quantity caption changes to **Bin Empty** with the quantity value shown as a red 0.

- The displayed 0 is display-only — the recorded tank amount and quantity applied are not changed, and the real value returns as soon as the alarm clears.
- The alarm clears immediately when the bin refills or the master switch is turned off. Tap the alarm button to silence the buzzer.

> **Tip:** Use a plain digital pin for the bin sensor. Inputs with analog conditioning on the board (such as the pressure input) load the internal pull-up and may not read reliably as a digital input.

Controls

Control	Action
Cancel	Discard changes
Save	Save settings

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21. Modules – Relay Pins

Menu: Modules > Relay Pins

Assigns the physical GPIO pins to each of the 16 relay outputs on the module. Pin values range from 0–50. A value of 255 (displayed as "–") means the relay pin is not assigned.

- **Low Relays (1–8):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.
- **High Relays (9–16):** Labeled "Onboard Relays" or "Remote Relays" depending on module config.

Press **Rescan** to clear all relay pin assignments.

> **Tip: Load preset** in Modules > Config fills in the relay pins for a standard board.

Controls

Control	Action
Cancel	Discard changes
Save	Save settings

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22. Modules – Valves

Menu: Modules > Valves

Selects the wiring mode for the valve or motor driver.

Option	Description
2-Wire	Motorized or latching valve — the driver actively powers the valve in both directions to open and close it.
3-Wire	One switched signal wire, one power wire and one ground wire. Compatible with standard solenoid valves and motorized ball valves that manage open/close internally. Default for most installations.

Select the mode that matches how your valve or motor driver is wired to the module.

Controls

Control	Action
Cancel	Discard changes
Save	Save settings

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23. Rate Map

Menu: File > Rate Map

The Rate Map enables variable rate application by defining geographic zones with different target rates. It integrates with AgOpenGPS for GPS-referenced mapping.

Prescriptions Tab

Manages prescription shapefiles for the current field. The active prescription determines which zone rates are used during application.

Control	Action
Field	Shows the field name assigned to the current job
Prescription list	Lists all prescription files available for the current field. Select one to make it active. **Zones.shp** is the default and cannot be deleted.
New	Create a new empty prescription file with a given name
Copy	Copy the selected prescription to a new name. The copy becomes the active prescription.
Import	Open the prescription import dialog — imports a shapefile or an XML prescription. See Importing a Prescription below.
Export	Export the selected prescription shapefile to a chosen location
Delete	Delete the selected prescription file. Zones.shp cannot be deleted.

> **Note:** Selecting a prescription from the list immediately makes it the active prescription for the current job. The map reloads to show the selected zones.

Importing a Prescription

The Import button opens a two-step import dialog:

Step 1 — Import Data

Control	Description
Shape file / XML	Choose the source format: a prescription **shapefile** (.shp) or an **XML** prescription file
Import	Select the file and read it in
Mapping grid	For shapefiles, match each product (A–E) to the attribute column in the shapefile that holds its rate. For XML files, the layer names found in the file are shown.
Default column	Fallback rate per product (shapefile import only). Zones missing the mapped attribute receive this rate instead of 0. Tap the cell to enter a value; leave it at 0 for no fallback.
Product	The primary product (A–E) used when adjusting zones in Step 2. Selected automatically from the imported data; change it if needed.
Name	File name for the saved prescription

Step 2 — Adjust Zones (optional)

Grid-style prescriptions can contain thousands of tiny zones. Adjust consolidates them into fewer, larger zones that are easier to work with and apply.

Setting	Description
Max Zones	Maximum number of distinct rate zones to create
Min zone size	Zones smaller than this area (Ac/Ha) are merged into their neighbours
Min zone step	Minimum rate difference between zones — rates closer together than this are combined
Adjust	Run the consolidation and report how many zones were created

Press **Save** to save the prescription for the current field. It becomes the active prescription. If the imported data has different product names, RC offers to update the job's product names to match.

Zones Tab

Create and edit rate zones within the active prescription.

Create zones by clicking the **New** button, then clicking at least **3 points** on the map to define the boundary.

- With **overlapping zones**, the **last zone added** takes priority.
- When variable rate is enabled but **no zone exists**, the **base rate** is used.
- Switching to another tab ends an active **New**/**Edit** — unsaved zone changes are discarded.

Button	Action
New	Create a new rate zone
Edit	Edit zone properties (name, rate per product, color)
Delete	Delete corner point (while creating) or delete zone (while editing)
Cancel	Cancel current edit
Save	Save current edit
Zone List	Open the Zone List — view all zones with their areas, product rates (A–E), and colors; tap a rate cell to change it; select/deselect zones; delete zones; print the list. To set one rate on many zones at once, check the wanted zones, choose the product (A–E), enter the rate and press **Apply** — then Save or Cancel as usual.

Data Tab

Control	Action
Record count	Number of applied rate data points recorded
A / B / C / D / E	Select which product's applied rate to display on the map
Record	Start/stop recording applied rates
Delete	Delete all recorded applied rate data

VR Tab — Variable Rate Look-Ahead

Configures how far ahead of a zone boundary the application rate begins adjusting, so the rate matches the target as soon as the machine enters the new zone. Compensates for applicator response lag.

Setting	Description
Look-Ahead Time (per product)	Seconds before entering a zone boundary that rate adjustment begins. One value per product (up to 5).
Auto Tune	RC monitors look-ahead performance and automatically adjusts the value to improve zone entry accuracy.

Files Tab

Import and manage supporting data files for the current field. Use the file selector dropdowns to choose which file is active for each overlay layer.

Field Boundary

The field boundary defines the operated area and is used to clip automatically generated productivity zones. Select the active boundary from the **KML** dropdown. Import a boundary using the **Import Boundary** button — three file types are accepted:

- **AOG Field.kml** — the field KML from the AgOpenGPS Fields folder; the boundary polygon is extracted automatically.
- **AOG Boundary.txt** — AgOpenGPS boundary file; easting/northing coordinates are converted to lat/lon using the field origin.
- **Any KML (*.kml)** — any KML file containing a polygon boundary.

> **Tip:** Field files are typically found in:
>
> - **AgOpenGPS:** Documents\AgOpenGPS\Fields\[Field Name]\ — contains `Field.kml` and `Boundary.txt`
> - **TWOL:** Documents\TWOL\Fields\[Field Name]\ — contains `Boundary.txt`

Yield, EC, and Elevation Data

Control	Action
Yield dropdown	Select the active yield data file for the overlay and zone creation
Import Yield	Import a yield data CSV file. If the file contains elevation data, RC will offer to extract it to a separate elevation file.
Delete Yield	Remove the selected yield data file
EC dropdown	Select the active EC (electrical conductivity) data file
Import EC	Import an EC data CSV file (Lat, Lon, EC)
Delete EC	Remove the selected EC data file
Elevation dropdown	Select the active elevation data file
Import Elevation	Import an elevation data CSV file (Lat, Lon, Elevation)
Restore Elevation	Restore the previous elevation file from the automatic backup
Delete Elevation	Remove the current elevation data file

Create Zones

The **Create Zones** button opens the productivity zone creation dialog. Zones are generated automatically from yield history, soil EC, elevation data, or any combination. Clear existing zones before generating if you want to replace them.

Setting	Description
Yield files (checklist)	Check one or more yield data files to include. Multiple years are blended by their assigned weight.
EC	Include the active EC layer in zone generation
Elevation	Include the active elevation layer in zone generation
Weights	Percentage weight for each active layer. All weights must sum to 100 before Build is enabled.
Productivity levels	Number of productivity classes to create (e.g. 3 = Low / Med / High). Each class may produce more than one zone polygon if the field has non-contiguous areas of the same productivity level.
Min zone size	Minimum zone area in hectares (or acres). Zone polygons below this size are discarded. Increase to reduce the number of small fragments.
Build	Generate zones using the selected layers and settings. The field boundary (set in the KML dropdown) clips zones to the operated area.

> **Tip:** Set the field boundary KML before generating zones. Without a boundary, zones are clipped to the bounding box of the data files, which may not match the actual field edge.

Display Toggle Buttons (right panel)

These buttons highlight green when active. Click to toggle on or off.

Button	Action
Sat. View	Toggle satellite background image
Target Zones	Toggle rate zone boundaries display
Applied Rates	Toggle applied rate colour overlay
KML Files	Toggle KML file overlay
Yield Data	Toggle yield data colour overlay
EC	Toggle soil EC colour overlay
Elev.	Toggle elevation colour overlay

Bottom Controls

Button	Action
Minimize	Shrink the map to a small window. Drag it by its title bar to reposition.
Centre	Centre the map on the current GPS position
Help	Open the help page
OK	Close the Rate Map

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24. Rate Graph

Menu: Help > Rate Graph (opens as a separate window)

Displays a real-time chart of the PID control loop for a product. Used for tuning Gain and Integral settings.

Displayed Data

Element	Description
Target UPM line	The desired flow rate setpoint
Applied UPM line	The measured actual flow rate
Error %	Current deviation from target
PWM value	Current PWM output (0–255)
Frequency (Hz)	Current flow sensor frequency

Chart Controls

Control	Description
Auto	Auto-scale the Y-axis to fit the data
Up	Increase Y-axis range by 1.5×
Down	Decrease Y-axis range by 0.67×

The chart maintains the last 30 data points. Click the graph to select which product to display.

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25. Scale / Tank Display

Menu: Products > Settings > Scale Weight (enable checkbox to show)

Displays a floating window showing tank/scale data for a product. Each product has its own scale display window.

Display Modes

Left-click on the icon to cycle through four modes:

Mode	Description
Current Weight	Net weight from the scale sensor (after tare)
Applied Weight	Difference between starting weight and current weight
Area Covered	Area worked since the last reset
Application Rate	Applied weight divided by area covered

Controls

Action	Result
Left-click on icon	Cycles the display mode
Left-click on value	Resets the starting weight and acres reference point
Right-click on value	Resets the tare (zero calibration)
Drag the background (either mouse button), or right-click and drag the icon	Repositions the window. The value can't be used to drag, since clicking it resets.

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26. Language Selection

Menu: File > Language

Changes the display language of the application.

Supported Languages

- English
- Deutsch (German)
- Hungarian
- Nederlands (Dutch)
- Polish
- Russian
- French
- Lithuanian

> **Note:** A restart of the application is required for the language change to take effect.

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27. Color Customization

Menu: File > Color

Customizes the foreground and background colors of the application interface.

Preset Color Schemes

Three built-in color presets are available. Select one to apply it immediately.

Custom Color

Select **Custom Color** to open the color picker:

1. Click in the **color gradient panel** to select a hue and saturation.
2. The horizontal position controls **hue**.
3. The vertical position controls **brightness/saturation**.
4. Separate foreground and background colors can be selected.

Color preferences are saved and applied immediately across all screens.

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28. Help – Logs & Diagnostics

Menu: Help (main menu)

Provides access to application logs, module diagnostics, and system information.

Log Files

Log	Description
Activity Log	Records application events and UDP/CAN communication activity in real time
Error Log	Records application errors and exceptions
Ethernet Log	Records raw UDP packet traffic — useful for diagnosing communication issues

Press **Start/Pause** to freeze or resume live log updates.

Activity Log — Module Status Events

When a module connects or its status changes, RC logs the following to the Activity Log automatically:

Log Entry	Description
Module X, Ethernet connected: True/False	Whether the module's Ethernet port is connected
Module X, Wifi Strength: 0–3	ESP32 WiFi signal level — 0 = not connected, 1 = weak (RSSI < -80), 2 = moderate (RSSI < -70), 3 = good (RSSI < -65)
Module X, Valves are 2 Wire / 3 Wire	Valve wiring mode reported by the module
Module X, Pin Configuration correct / not correct	Whether the module's pin configuration is valid

These entries are written once on first connect and again whenever the status changes.

Controls

Control	Action
Start/Pause log	Freeze or resume live log updates
Reset logs	Delete all log files after confirmation
PID Logging	Toggle PID diagnostic logging. While on (green), the module streams per-loop rate control data (target, applied rate, PWM, error) which is saved to a CSV file in the `PIDLogs` folder inside the data folder — useful for tuning and diagnosing rate control problems. The setting is remembered between sessions.
Open rate graph	Display the rate graph
Check for updates	Check for a newer version of the application
Open user manual	Open the user manual
Navigate modules left	View previous module diagnostics
Navigate modules right	View next module diagnostics

Module Diagnostics

Display	Description
Module ID	ID of the module currently being viewed (use left/right arrows to switch)
Firmware Version	Firmware version running on the selected module
Elapsed Time	Seconds since last module data was received

System Information

Display	Description
IP / Subnet	Current Ethernet subnet in use
Application Version	Installed RC software version and date
Current File	Name of the active profile file
GPS Coordinates	Live latitude and longitude from AgOpenGPS

External Links

Links to the project repositories are available from this screen:

- AOG RC GitHub repository
- PCB Setup GitHub
- Rate Control PCBs GitHub

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29. Tips & Tricks

Main Screen Navigation

- **To reposition the main screen**, press and hold (or right-click) in the middle of the screen, then drag it to the desired location. The product and fan icons only drag with the right mouse button, since a left click on them selects the product.
- Use the **minimize button** on the main form to reduce the display to rate only. RC remembers this — if it was minimized when closed, it starts minimized next time.

Clickable Display Elements

- **Click on target rate label** to switch between the target rate and the alternate rate (a percentage of the target). When variable rate is enabled, the label shows the VR target rate and is not clickable.
- **Click on quantity applied label** to switch between quantity applied and quantity remaining.
- **Click on area applied label** to switch between area applied and area remaining.
- **Click on area applied value** to reset the area counter.
- **Click on the tank bar graph** to reset the tank quantity.
- Click a **product icon** at the top of the screen to select which product's rates and totals are displayed. The rate graph will reflect the currently selected product.

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30. Troubleshooting

No Relays Activating

1. Is the module Config set correctly for board type and relay type?
2. Has the subnet been set in the app and uploaded to the module?
3. Have section count and widths been set?
4. Is Auto mode on?
5. Is a section assigned to each relay?
6. Is a switch assigned to each section?
7. If Auto is on, have sensor counts/unit and base rate been set?
8. If Work Switch is enabled (Menu > Switches), the master will only activate when the work switch reads on — check the switch's wiring, pin assignment, and the **Invert Work Switch** setting (Modules > Pins) against whether the sensor is normally-open or normally-closed.

No UPM (No Flow Reading)

1. Check ground speed.
2. Check working width.
3. Check sensor control settings.
4. Is AgOpenGPS connected?
5. Is the module connected?
6. Check sensor counts/unit.
7. Check base rate.
8. Have control values been sent to the module?
9. Check control type — is it a valve or motor?
10. Is the connector fully plugged in?
11. Is the module running the latest firmware?
12. Is Auto mode on?
13. If Work Switch is enabled, is it reading on? A wrong **Invert Work Switch** setting (Modules > Pins) will hold the master off with no flow, even though everything else is configured correctly.

Document Applied Manual Rate (Mode 5)

1. Don't assign a real module to a product set to Mode 5 — the module's real sensor data will conflict with the manually entered value.
2. Enter the flow value in units per minute, not per acre/hectare; there is no rate alarm in this mode to catch a units mistake.
3. Applied rate will rise and fall with ground speed even though the flow input is constant — this is expected for a fixed-metering implement.
4. A ground speed source is still required — GPS (via AgOpenGPS), a wheel-speed sensor module, or a manually entered/simulated speed (Menu > Speed) all work; with no speed source active, quantity won't accrue even though the flow value is set.
5. Applied totals reading consistently high or low on a multi-dispenser machine usually means the section count doesn't match the number of dispensers. The entered value must be the whole machine's output

with all sections on, and there must be one section per dispenser with its width set to the ground that dispenser covers — RC scales the flow by section width and has no sensor to check it against.

AOG Status Light Not Green

1. The AOG light reflects whether AgOpenGPS is sending GPS or AutoSteer position data over the network — RateController does not need to send anything for it to turn green.
2. AgOpenGPS only sends that data once it has an active position source: a real GPS fix, or its built-in GPS Simulator turned on. No fix and no simulator means the light stays red, even if RateController and AgOpenGPS are both running correctly.
3. If your AutoSteer hardware also supplies GPS (common on combined GPS/IMU/steer boards), unplugging it removes the position source too — the AOG light will drop even though AutoSteer itself isn't what RateController is waiting on.
4. To test the connection without field hardware, turn on AgOpenGPS's GPS Simulator — the light should turn green the same way it would with a real fix.

No Rate Adjustment

1. In manual mode, the master switch must be on.
2. Is minimum PWM too low?

Motor Won't Shut Off

1. Minimum PWM should be set to 0.
2. Is master override enabled?

Updating Module Firmware

Board firmware is updated with **PCBsetup**, a separate program from RateController. RateController can stay open while a board is updated.

1. Open PCBsetup and press **Check for Updates** to download the latest firmware.
2. In **Application**, choose the board: **Teensy Rate**, **Nano Rate** or **ESP32 Rate**.
3. **Over USB (any board):** connect the board to the PC by USB. On the **Serial/USB** tab, choose its port, press **Connect**, then **Firmware**, and upload.
    - **Nano:** tick **Use Old Bootloader** for older Nanos. If the upload fails, switch the setting and try again. You may need to unplug the Nano from its Ethernet shield while uploading.
4. **Without USB:**
    - **Teensy:** on PCBsetup's **Ethernet** tab, press **Firmware** and enter the board's **Module ID**. This only works if the firmware already on the Teensy supports network updates.
    - **ESP32:** open the board's web page (see Module WiFi (ESP32)) and press **Update Firmware**. Choose `RC_ESP32.ino.bin` from PCBsetup's downloaded firmware, or from the release's `ModulesHex.zip`.
5. Then follow After Updating Module Firmware below.

After Updating Module Firmware

1. Use the RateController version released with the firmware. An older RateController shows an updated board as not connected.
2. The board keeps its module ID (0–7) but its settings go back to defaults. Resend them: Modules > Config, choose the module, then Send.
3. If the board doesn't connect, resend the subnet (Modules > Communication).
4. A module that shows **Firmware update required** (Config page) or **Update firmware** (Boards page) still has old firmware. Update it (see Updating Module Firmware above); RateController doesn't use that module until then.
5. A board that comes back as **New** on the Boards page is assigned its module ID there again (see Setting Up a New Board).
6. A board that isn't listed on the Boards page at all may still have older firmware (e.g. a Nano). Update it (see Updating Module Firmware above).

Phone Page Won't Open

1. The phone and the tablet must be on the same network (File > Options > Phone).
2. Phone hotspot or router: the network must be set to **Private** on the tablet (Options > Phone > WiFi settings). Needed again for a new phone or a renamed hotspot.
3. Tablet hotspot: the status must show *On*. If it shows *On, firewall not allowed*, choose Current WiFi Network on Modules > Communication > WiFi and Save, then Tablet Hotspot and Save, and accept the Windows prompt.
4. If the address shown isn't on the phone's network, tap the QR code to show the next address.
5. Router: guest networks and AP / client isolation stop the phone from reaching the tablet.
6. *Not running, port 8080 may be in use*: another program on the tablet is using port 8080.

Product Restored After an Interrupted Calibration

This message shows at start when RateController closed during a calibration (power loss, a crash, or ended in Task Manager). The product's original application mode, enabled state and off-rate alarm have been put back, and the speed source restored. Check them before continuing; a calibration that wasn't saved has to be repeated.

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